Eric Min

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Education

Massachusetts Institute of Technology — Cambridge, MA *Expected 2030*
B.S. in Aerospace Engineering

Oregon Episcopal School — Tigard, OR *2022 – 2026*
High School Diploma, GPA: 4.0/4.0 (unweighted) • AP Scholar with Distinction • National Merit Scholar (2026)

Courses: Differential Equations, Linear Algebra, Multivariable Calculus, Advanced Statistics with R, Python III: Data Science, Algorithms, Machine Learning

Portland State University — Portland, OR *2024 – 2026*
GPA: 4.0/4.0 (unweighted) • concurrent enrollment while in high school

Courses: Discrete Mathematics (MTH356), Introduction to Group Theory and Applications (MTH344), Optimization for Data Science (MTH566)

Research & Projects

MIT Rocket Team — Liquid Propulsion & Controls (LP&C) Division, Fluids Subteam *2026 – present*

- Contributing to Project Osiris; develop piping and instrumentation diagrams (P&IDs) and verify systems meet safety requirements in close coordination with the propulsion group
- Perform ullage calculations and tank slosh simulations for liquid propellant tanks

Portland State Aerospace Society (PSAS) — Propulsion & Controls Intern *2024 – 2026*

- Wrote a Python script implementing Rao's method to automatically generate bell nozzle geometry optimized for maximum specific impulse (I_{sp})
- Modeled generated nozzles in CAD and evaluated performance for the roll control system using CFD simulation
- CAD-modeled structural mounting components to secure a gas tank during launch; validated designs through stress simulation to a factor of safety of 1.5

Regeneron International Science and Engineering Fair (ISEF) — Finalist *May 2025*
"Flow Control via Bio-Inspired Leading-Edge Tubercles for Enhanced Airfoil Performance"

- Parametrically CAD-modeled tubercled airfoils and simulated flow in ANSYS Fluent; validated results by wind-tunnel testing of 3D-printed models
- Quantified L/D gains as a function of tubercle geometry (amplitude and wavelength) across multiple angles of attack, achieving an L/D improvement of over 20%

The American Rocketry Challenge (TARC) — Team Lead, Hardware & Software *2022 – 2026*

- Led a team of 20+ to a National Finalist finish, 20th of 922 teams; invited to NASA Student Launch Initiative
- Designed, fabricated, and tested an active altitude control system using variable-angle flaps actuated by an onboard microcontroller, with custom PCB and CAD-modeled mechanism

Technical Skills

Engineering: SolidWorks, Fusion 360, Rhino/Grasshopper, ANSYS (Fluent, Mech.), OpenFOAM, MeshLab, OpenRocket

Programming & Data: Python (NumPy, pandas, Matplotlib, scikit-learn, OpenCV), R, MATLAB, Git, LaTeX

Hands-on: PCB design (KiCad), 3D printing (FDM, MSLA), wind-tunnel testing

Honors & Awards

Mathematics: AIME Qualifier, 5x (2022–2026); AMC 10B: 136.5/150, Distinguished Honor Roll, Top 1% (2023)

Athletics: MIT Varsity Fencing; Team USA Athlete, Cadet World Cup Team Bronze Medalist (Šamorín, Slovakia, 2024); 4x High School All-American & Academic First Team (2023–2026)

Additional

Teaching: Taught introductory python from scratch to OES middle schoolers (2024–2025); math competition tutor for OES Lower & Middle School (2024–2026)

Languages: English (native), Korean (fluent)